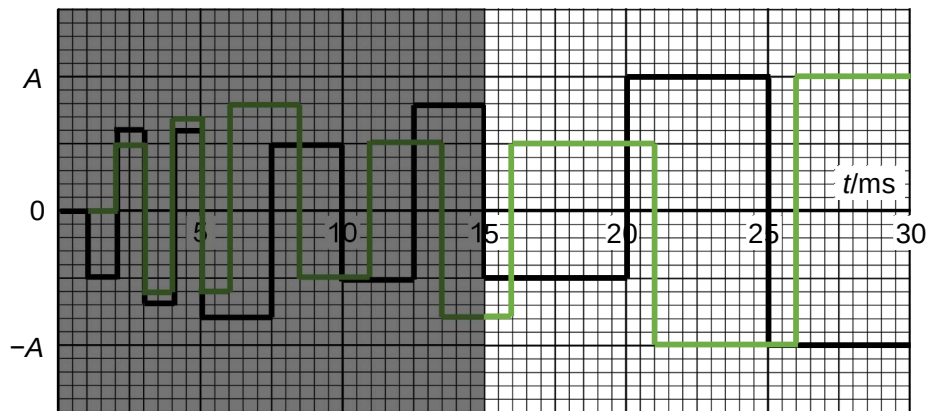


4(a) Two or more waves of the same kind meet at a point at the same time, displacement of the resultant wave is vector sum of the displacements of the individual waves at that point at that time. B1

4(b) constant phase difference of π (rad) B1
 same amplitude B1

4(c)(i) Amplitude



phase difference (shown by reflection in x-axis) B1
 time lag (shown by translation of 2 small squares to right) B1

Question	Marks
4(c)(ii) <u>longer durations of destructive interference</u> for noise waves of <u>longer periods</u> ; correspond to <u>lower frequencies</u>	M1 A1